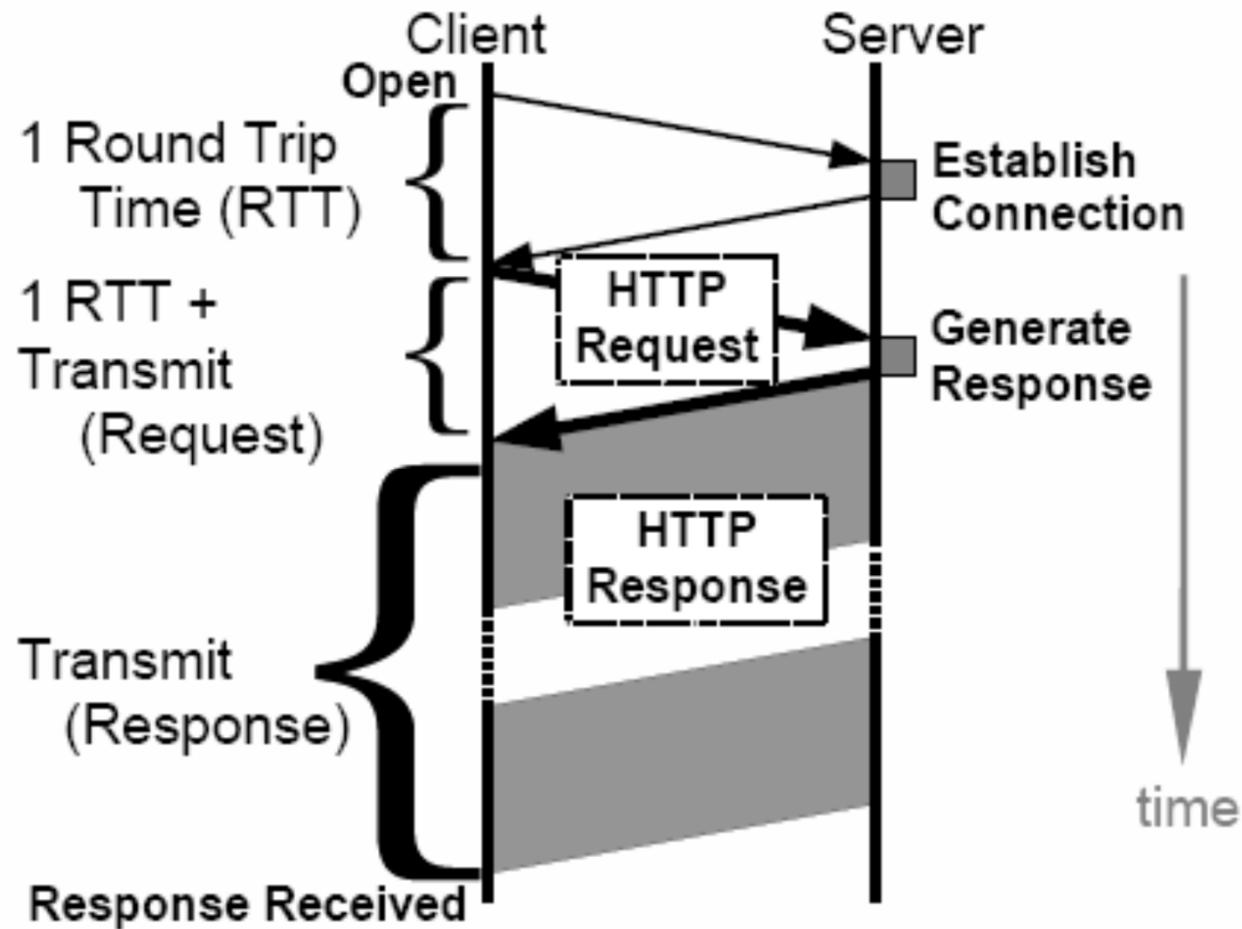


Steps on accessing a web site



- The time for DNS server to resolve the web server's name to IP address (if necessary).
 - Not shown in the diagram
- The time for the client to set up a TCP connection with the web server
- The time for the request transmitted from the client to the web server
- The time for the web server to parse the request and generate a web page
- The time for the response transmitted from the host to client
 - The time when all the information are received by the client
- The time for the client browser to render the page
 - Not shown in the diagram

How to reduce the response time

- Reduce the network latency (i.e., transmission delay)
 - Place servers closer to the clients
 - Could be significant
- Reduce the amount of information that have to be sent to the clients
 - Benefits the users that connect to the servers through low-bandwidth networks
 - For a given bandwidth, less information to be transmitted means less time is needed for transmission
- Reduce the load on the web server
 - Generate the response faster
- How to achieve these?
 - Proxy server, caching server
 - A proxy server normally caches web contents. But web contents can be cached on other types of servers as well.

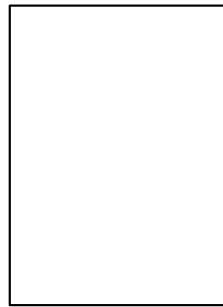
Web Cache

缓存

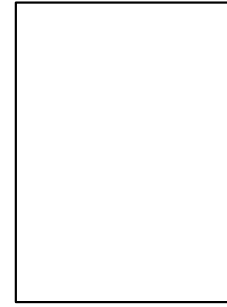
- A web cache usually sits between the web server and clients
 - Watches requests come by and saves copies of the responses for them.
 - When there is a coming request for the same URL resource, the web cache can retrieve the cached response and send it back to the client without asking for the web server to serve this request again.

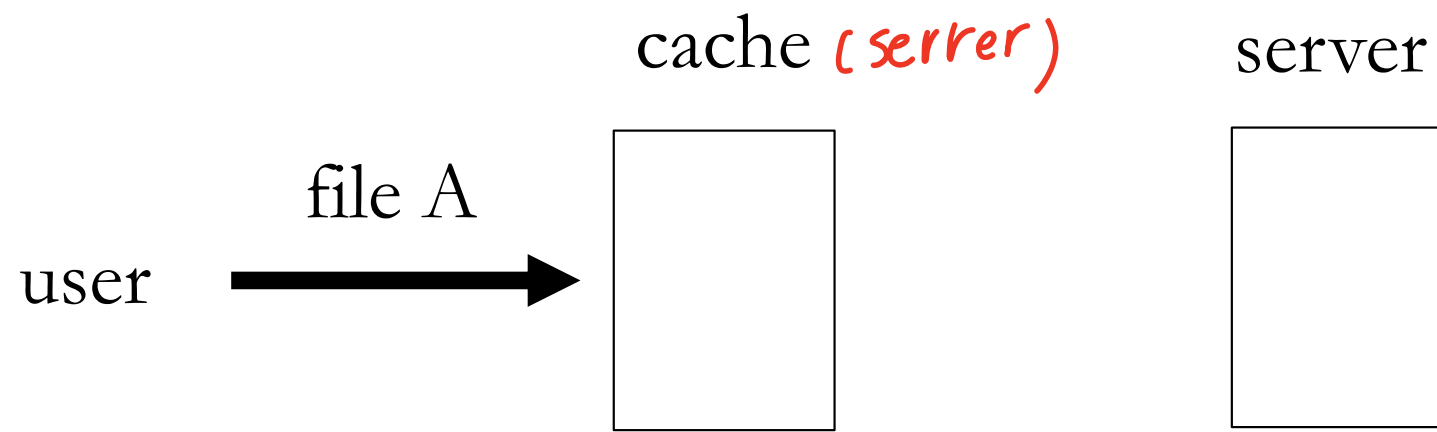
user

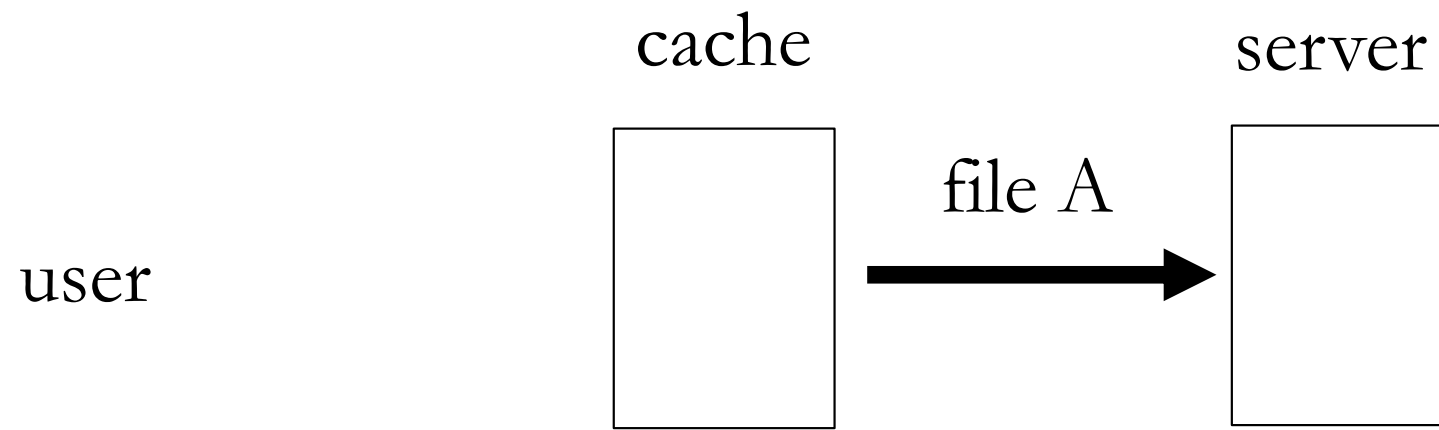
cache

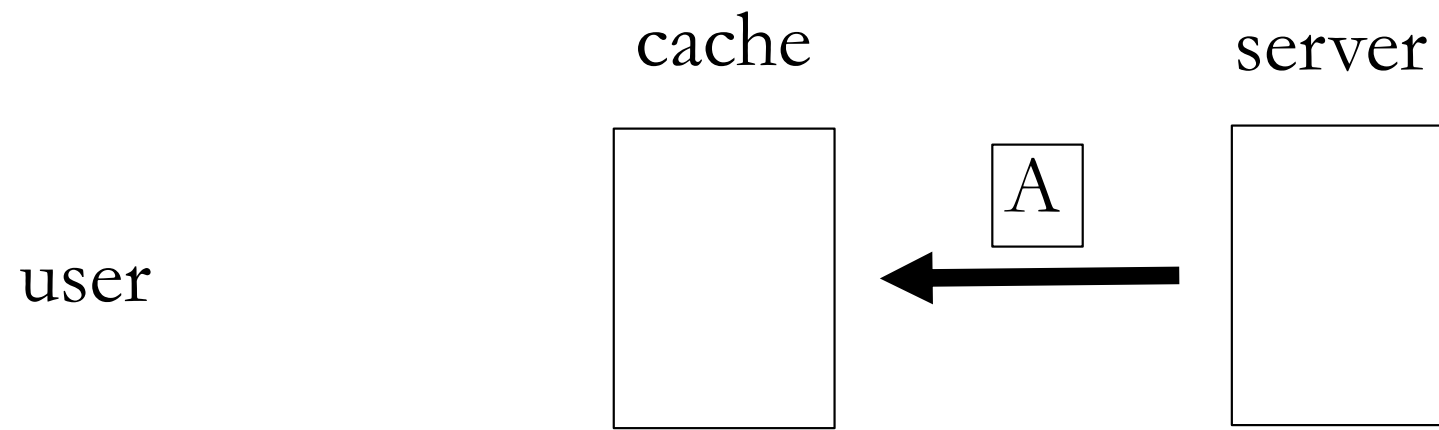


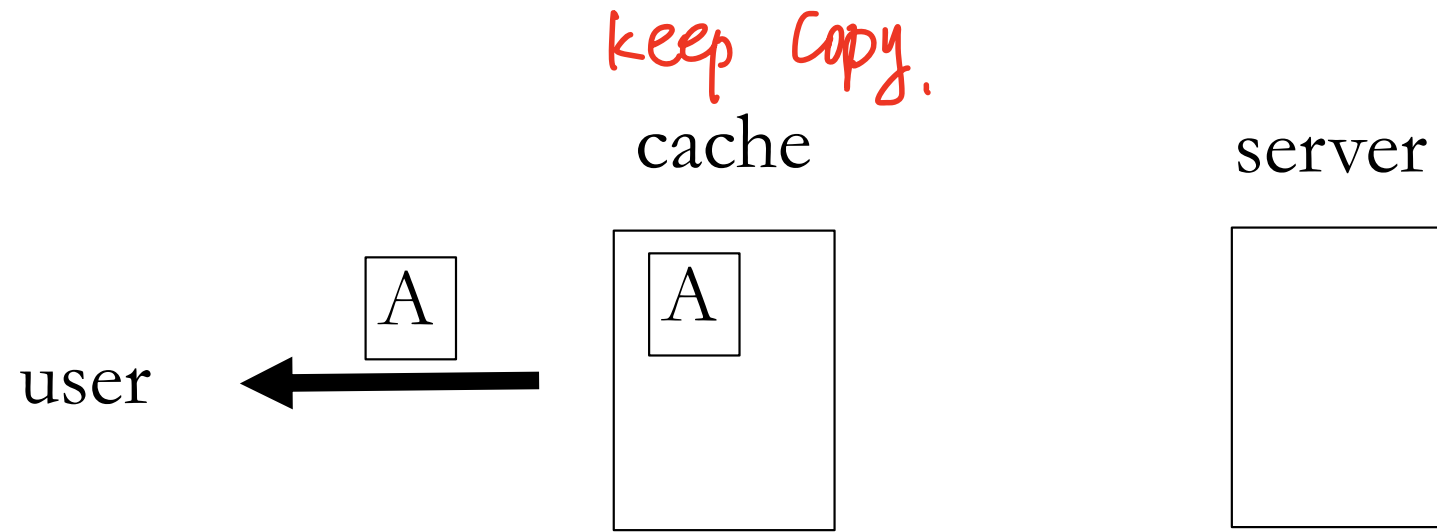
server

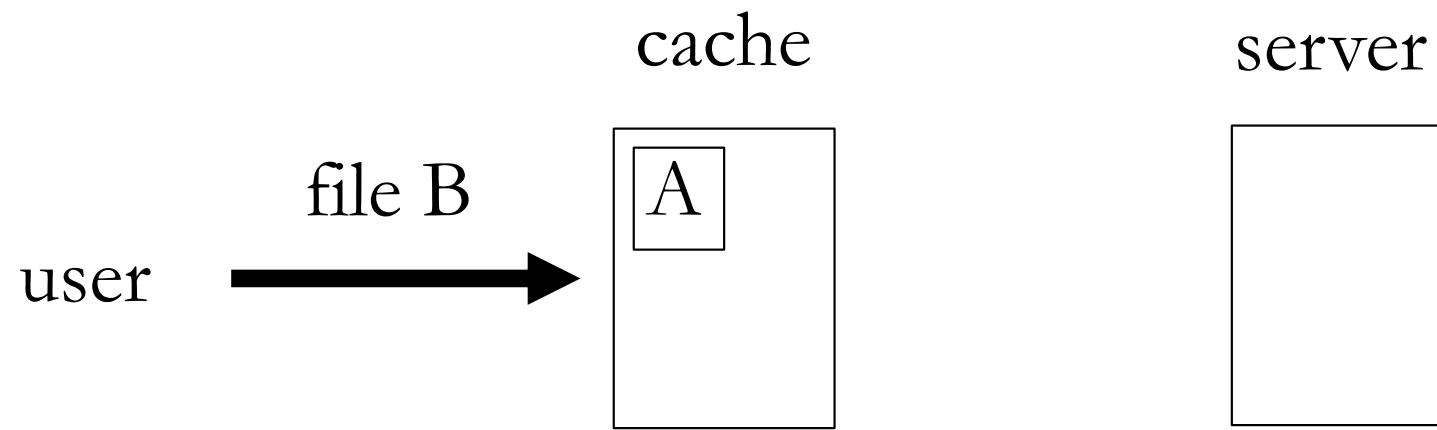


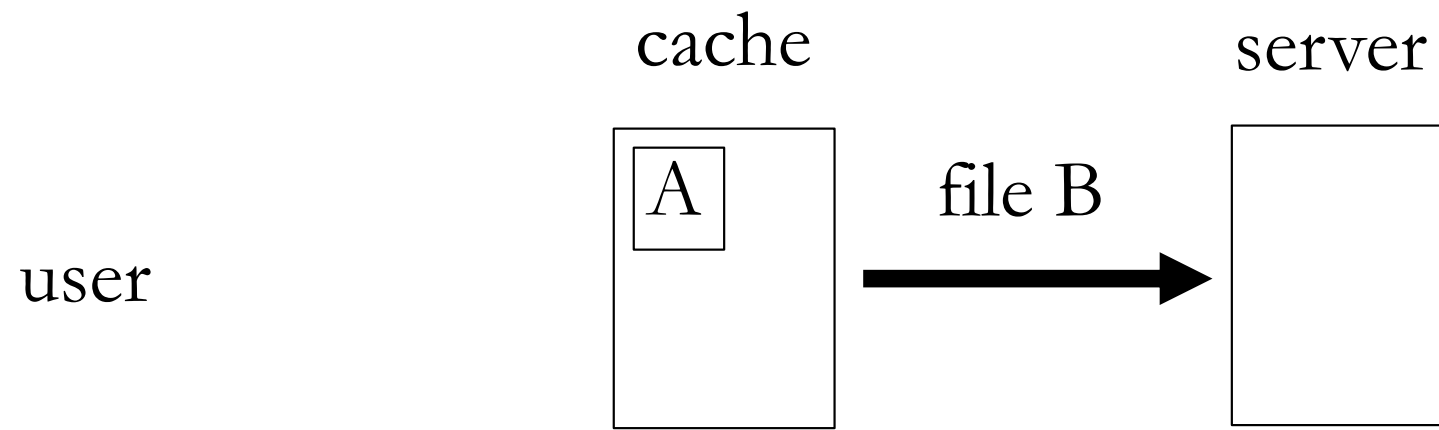


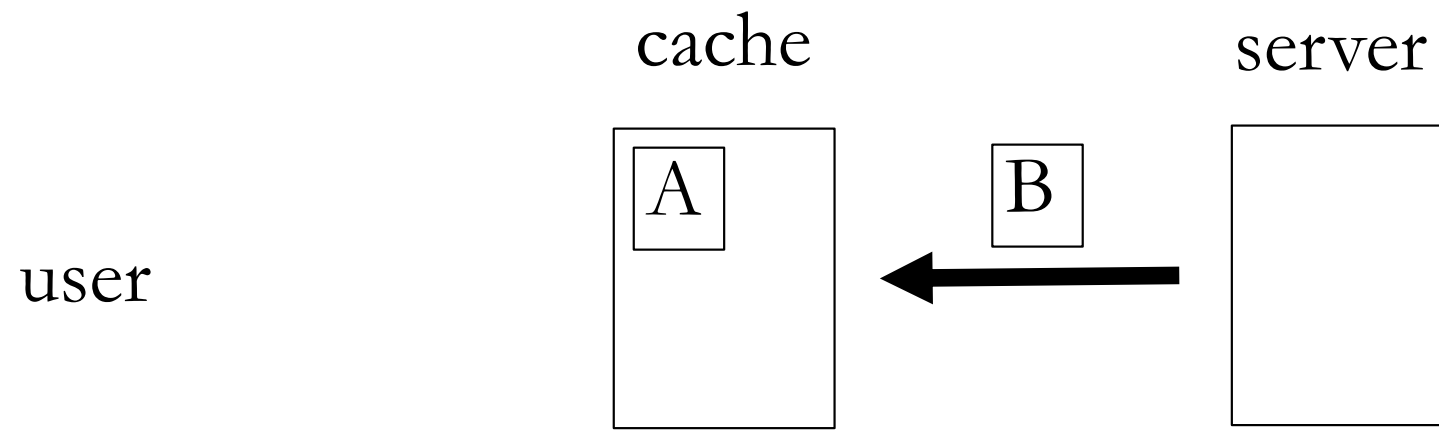


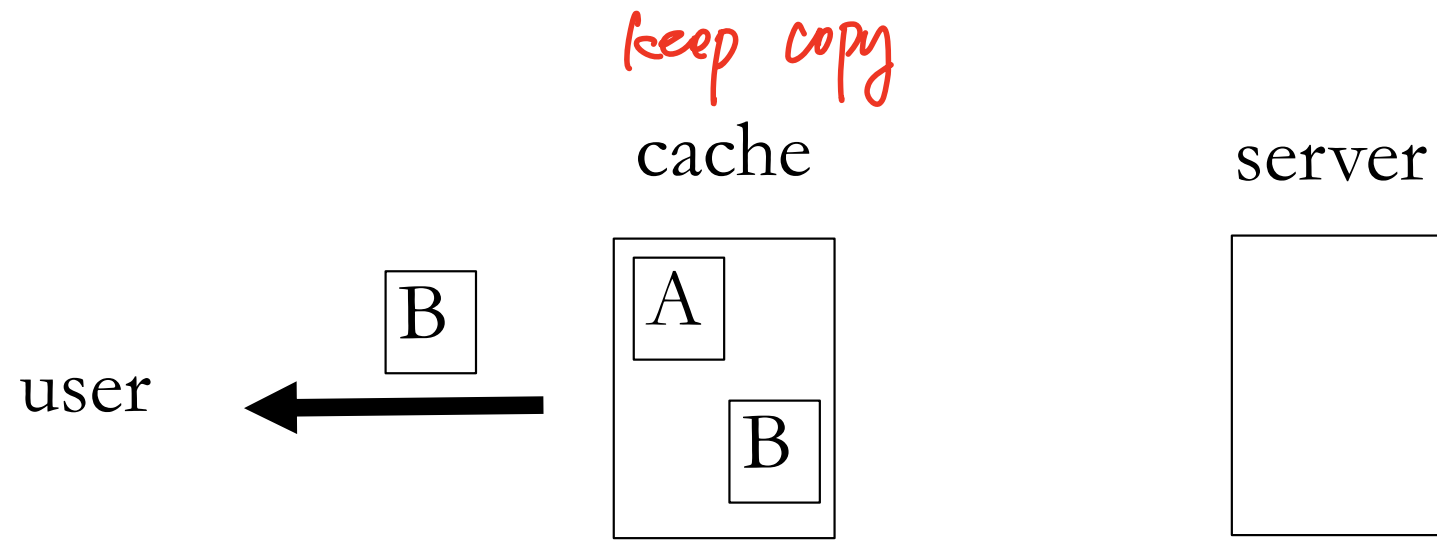


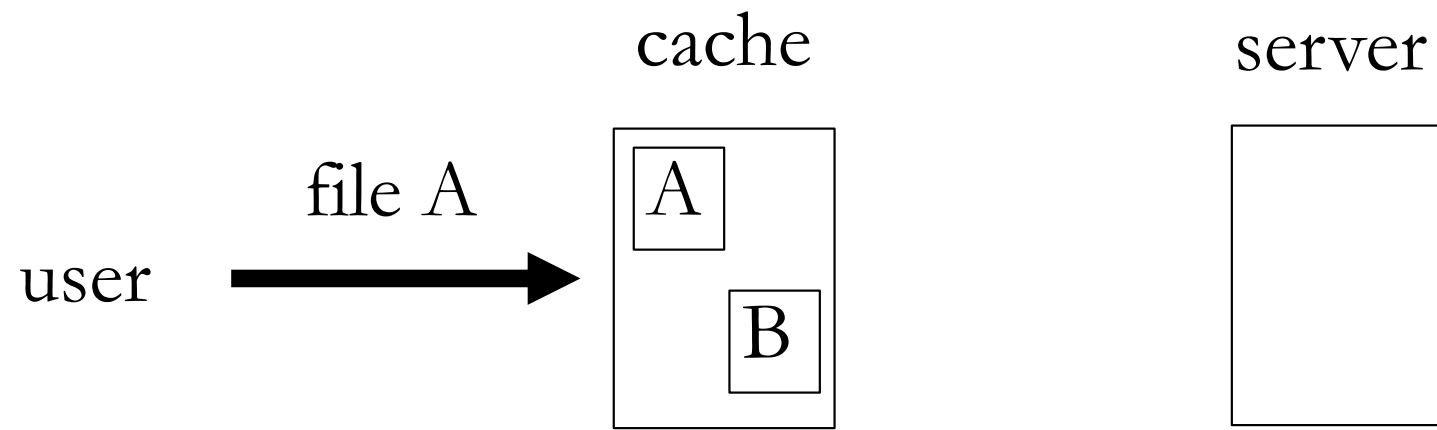


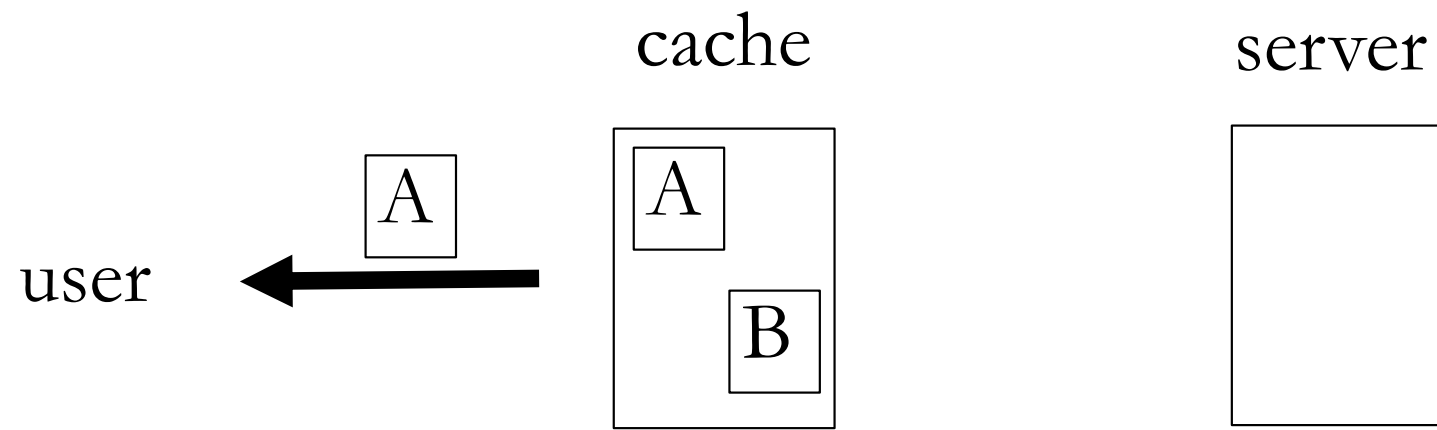












Advantages of Web Caching

- Reducing network latency
 - There is a distance between clients and web servers.
 - Caching web object locally can dramatically reduce transmitting latency

Table 5-1. *Latency and Response Time*

<i>Connection Speed</i>	<i>Latency</i>	<i>Response Time</i>
10 mbps	10 milliseconds	0.5 seconds
56 kbps	200 milliseconds	6 seconds
56 kbps	500 milliseconds	8.5 seconds



closer
far

web page



<https://pic.com/1.jpg>

<https://pic.com/2.jpg>

more than one connection.
several times

<https://pic.com/1.jpg>

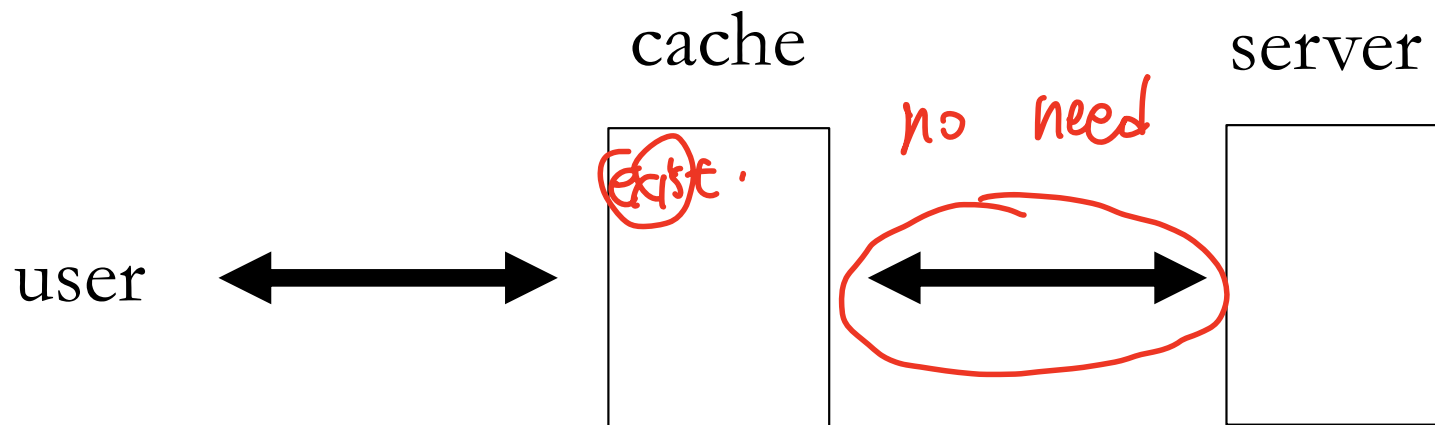
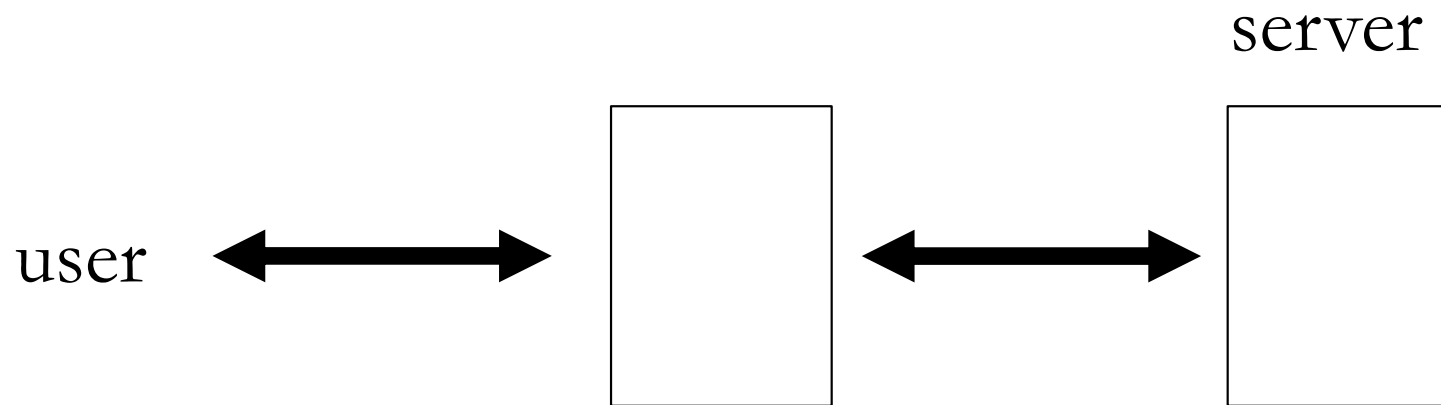


<https://pic.com/2.jpg>

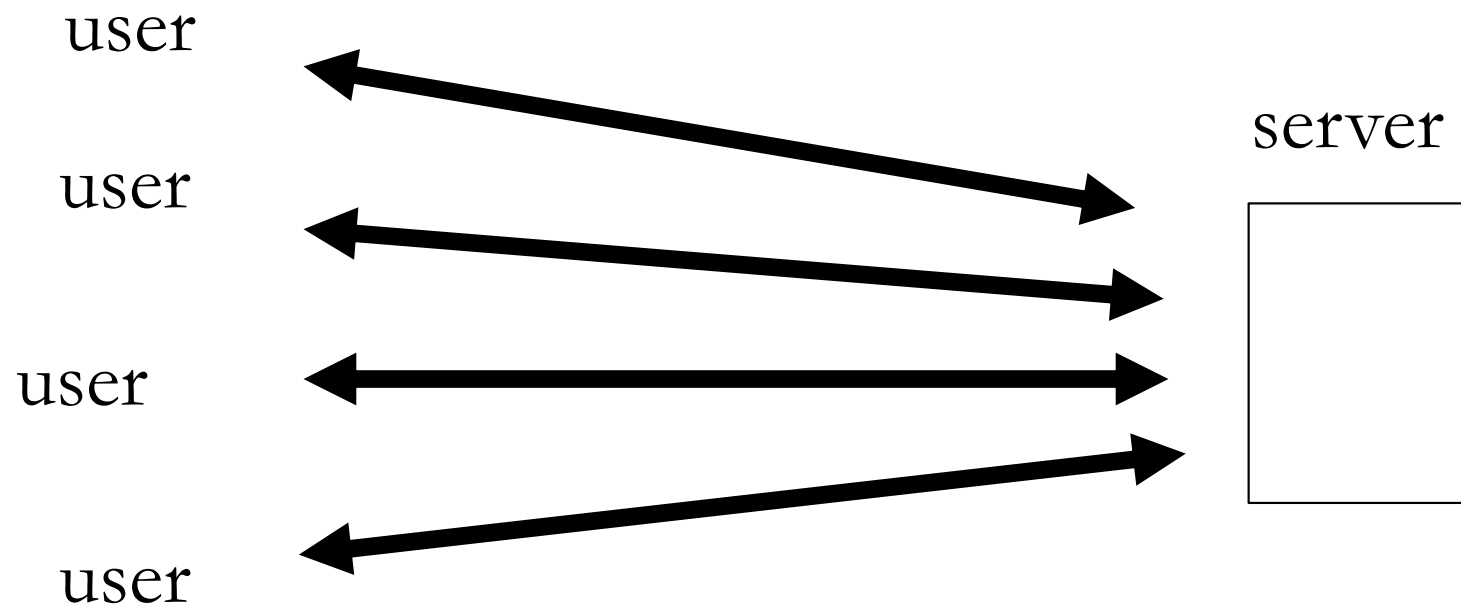


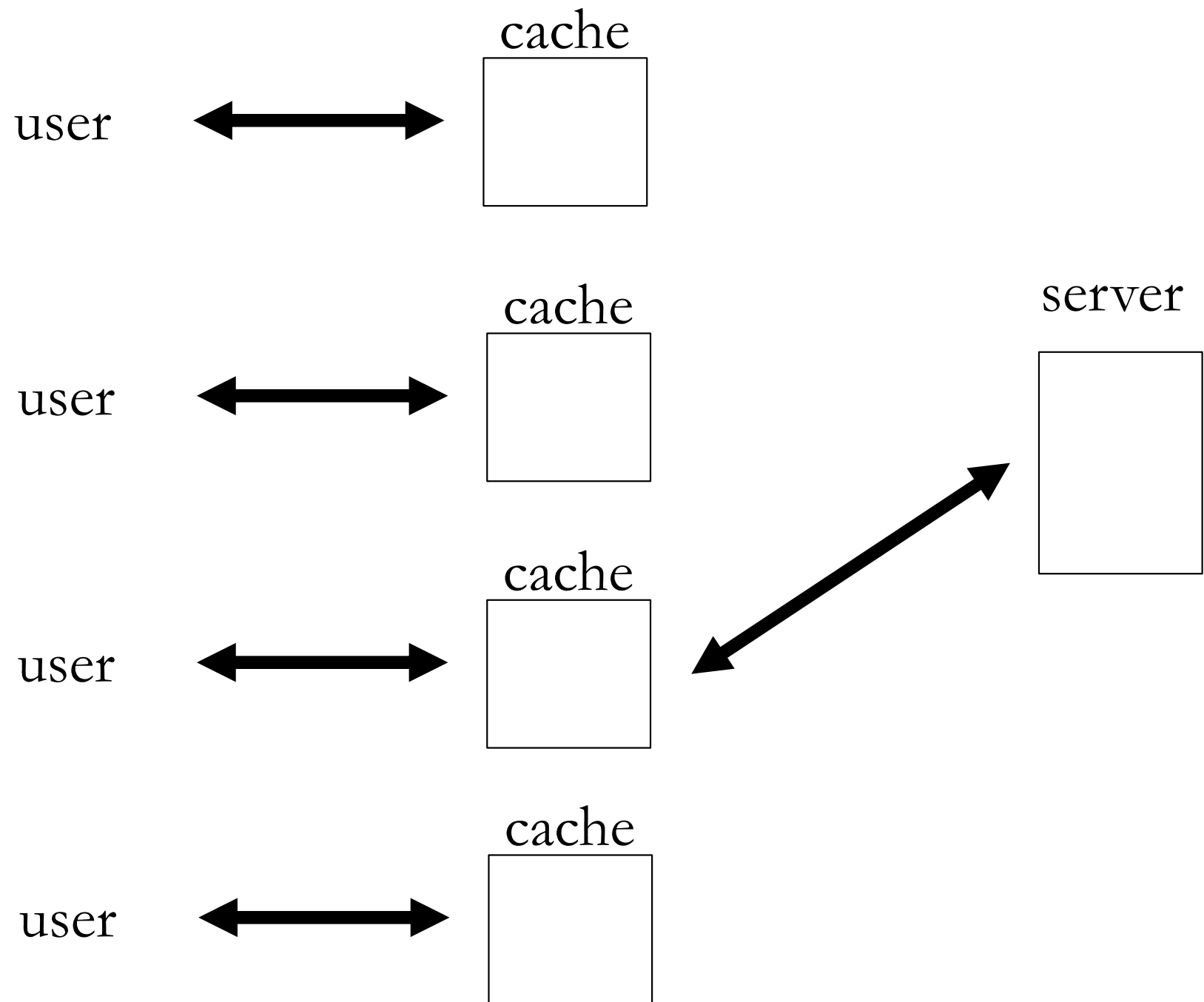
带宽

- Reducing bandwidth consumption
 - Cache the web objects close to the client side
 - Responses to the same requests do not need to be transferred again between the proxy server that caches the web objects and the origin web server.



- Reducing server load
 - Some of the requests do not need to be handled by the origin server.
 - As some requests can be served from proxy servers, the origin web server can direct more capability to deal with other requests.
 - Improve the response time for the non-cached contents.





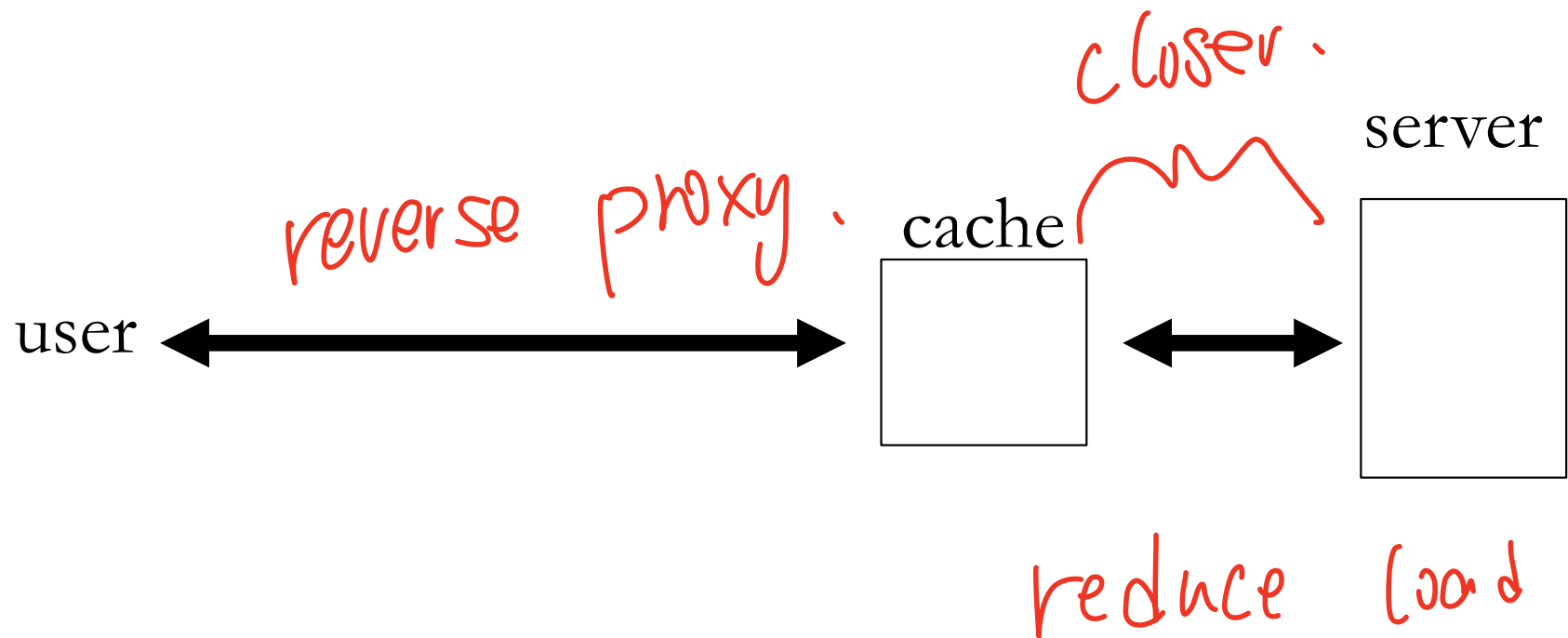
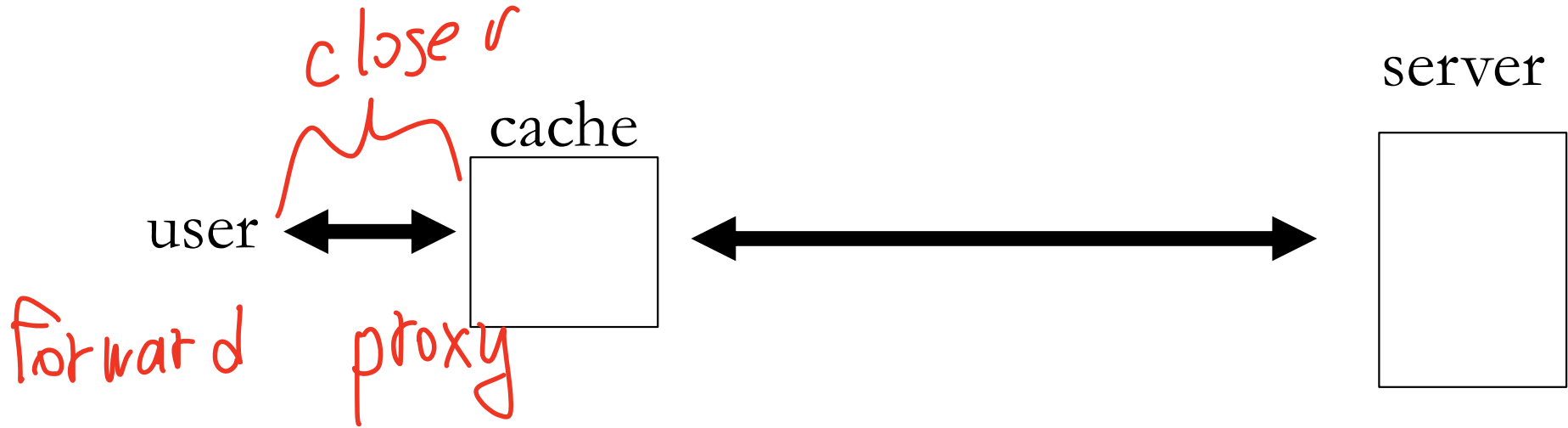
Categories of Web Caching

■ Browser

- Many web browsers support caching.
- When a web page is first requested, the browser saves the web page on the disk.
- If the page is requested again or the user clicks the “Back” button, the cached page might be used.
 - In some cases, the browser might need to check with the remote web server to see whether the cached page is still valid. If it is valid, the page in the local disk is used.
- Examples: Firefox, Chrome.

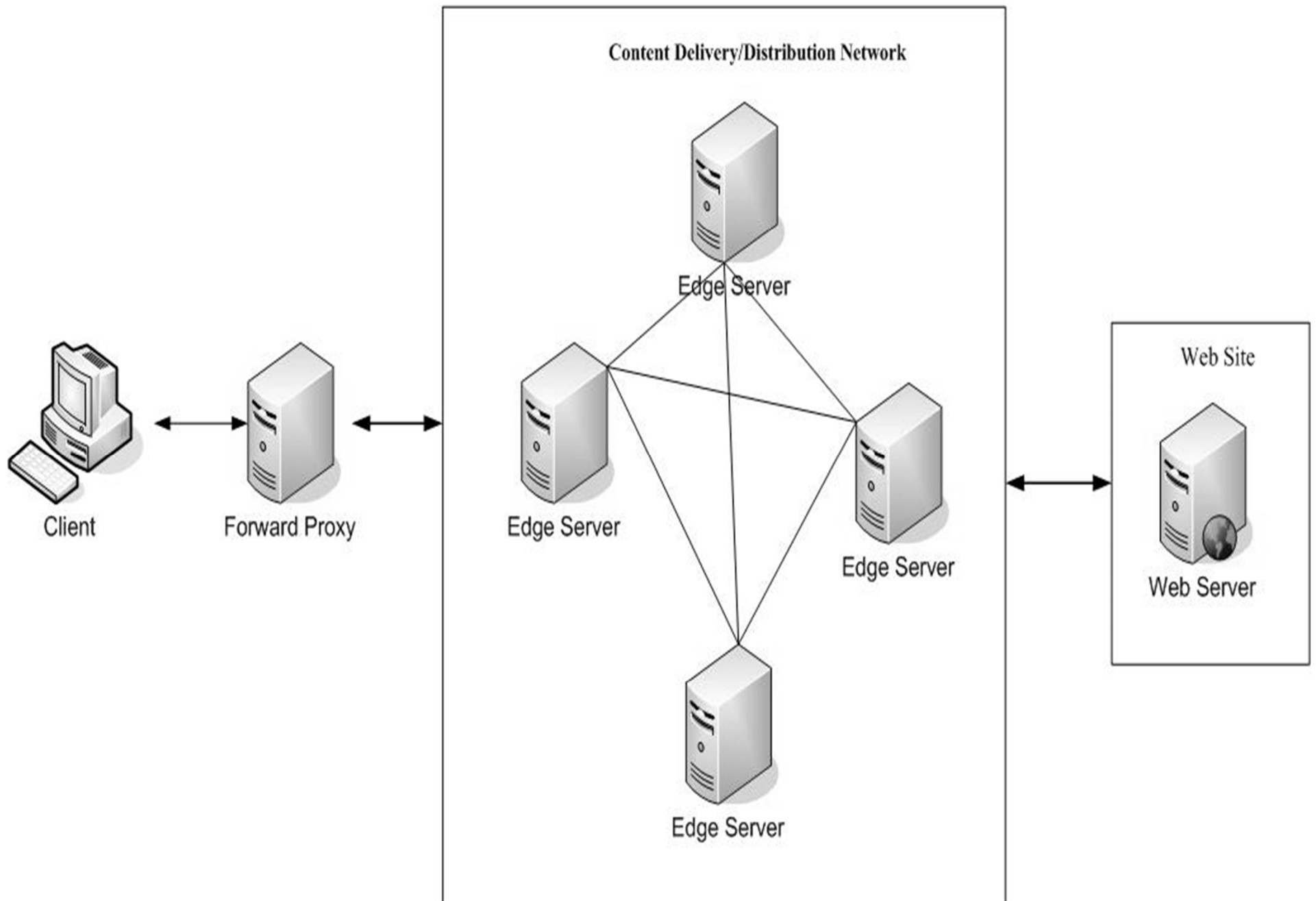
Proxy Server

- There are two types of proxy servers, i.e., forward proxy and reverse proxy
- A forward proxy is an intermediate system that enables a browser to connect to a remote network to which it normally does not have access. A forward proxy can also be used to cache data, reducing load on the networks between the forward proxy and the remote web server.
- A reverse proxy (http accelerator) is installed in front of web servers. All connections coming from the Internet addressed to one of the web servers are routed through the proxy server, which may either deal with the request itself or pass the request to the web server behind.



Content Delivery/Distribution Network (CDN)

- CDN is designed to place content caching servers at the edge of network around the world.
- It reduces the traffic congestion on the network and the latency perceived by users.
- The caching servers at the edge side of network are called edge servers which can cache HTML pages, images, audio and video files.
- Examples: Akamai, Digital Island



- When the client requests a web page, the request is firstly sent to a forward proxy.
- The forward proxy then decides which edge server to communicate with.
 - If none of the edge servers cache this page, the request is delivered to the web server.
 - Otherwise, one of the edge servers , which is closest to the client, will serve the request.
- Examples: Akamai EdgeSuite, IBM WebSphere Edge Server

Potential problems

- The end user see stale (e.g., old or out-of-date) content, compared to what is available on the origin server (i.e., fresh content).
out of date
- HTTP does not ensure strong consistency
- Caching tends to improve the response times only for cached responses that are subsequently requested (i.e., hits).
只对多次请求相同生效, 降
- Misses that are processed by a cache generally have decreased speed. Thus, a cache only benefits requests for content already stored in it.
低
延迟
- Caching is limited by a high rate of change of source data.
data change frequency
- Some responses cannot or should not be cached
不能缓存
- Web server will set and send the HTTP headers that determine cacheability

Dynamic Data Page

- Dynamic data pages are generated by applications according to the underlying frequently changing data sources such as databases and files.
- Dynamic data pages are usually generated on the demand of user's request. 波动性.
- Dynamic pages have two characteristics: high volatility and high variation. content varies
content change
 - It is volatile because not only it changes more frequently but also its change is unpredictable.
 - It is various because it is often customized or personalized from a set of content fragments.
- Compared to a static page, a dynamic data page is expensive to create.
 - more valuable to cache a dynamic data page

Proxy-based Dynamic Page Caching

- Proxy-based caching approaches can be classified into three broad categories
 - page-level caching
 - fragment-level caching
 - data-centric caching

} talk about
- In the page-level caching, the proxy server caches full-page outputs. 缓存整个页面
 - Many commercial products have applied this approach.
- Compared to no-caching, the page-level caching has the following advantages:
 - reduce the page generation delay
 - reduce the delay associated with packet filtering and other firewall-related delays
 - reduce the bandwidth required to transmit the content from the web server to the proxy cache server

- The page-level caching has the following limitations

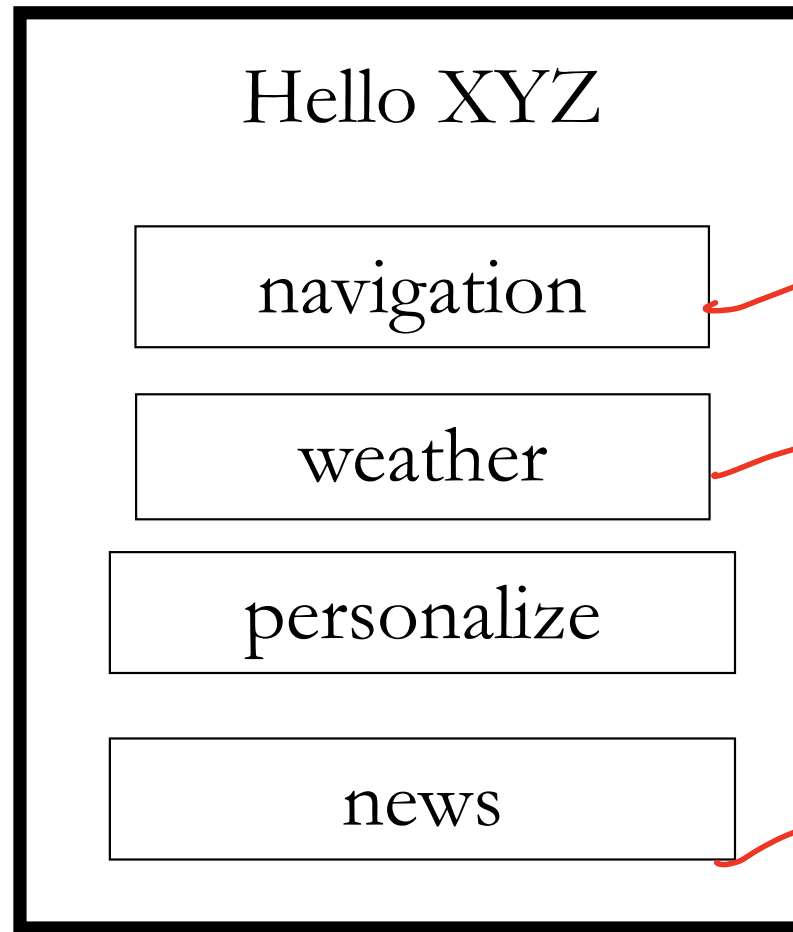
- There is often very little reusability of full HTML pages

很少有重用

- A personalised page displaying the name of the user at the top of the page: the cached page in the proxy is only reusable if the same user accesses to the same page

- Unnecessary invalidation

- If there is only one element of the page becomes invalid, then the whole page needs to be invalidated in the cache.

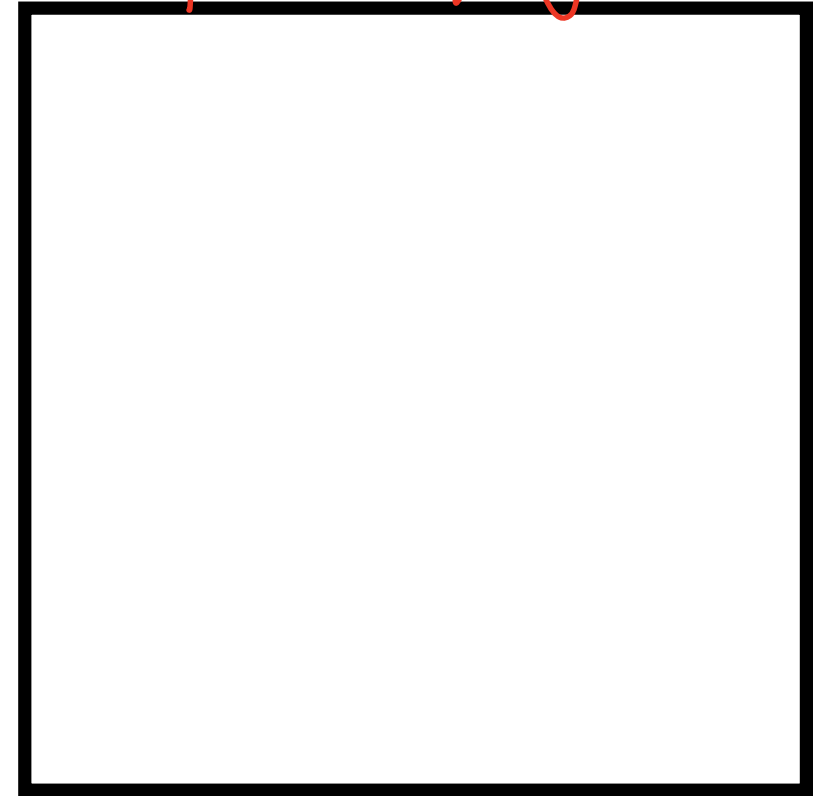
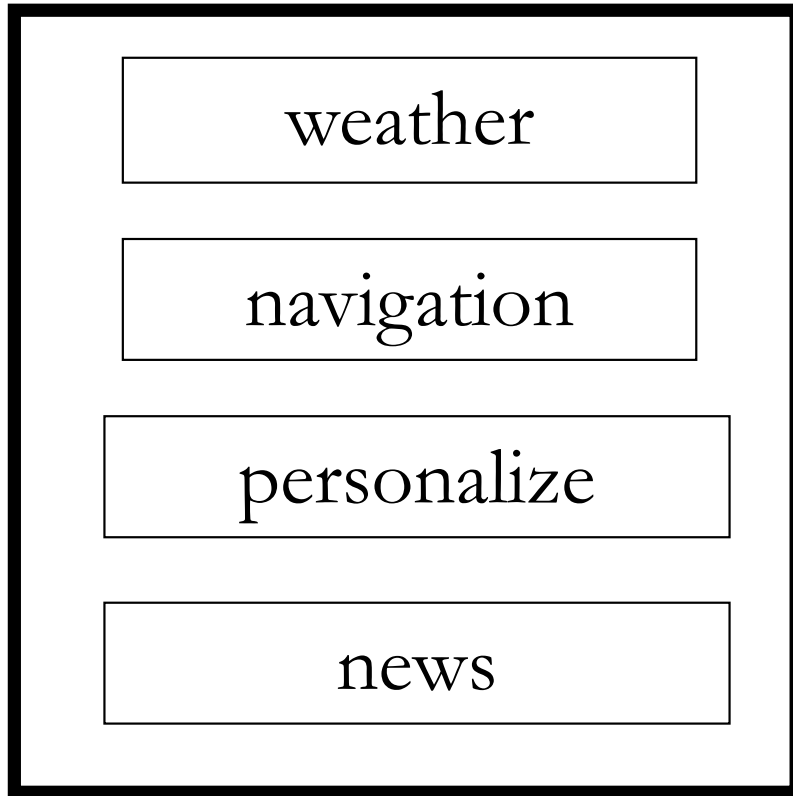


比较最新更新的时间，来决定是否重用

片段级 Fragment-level caching

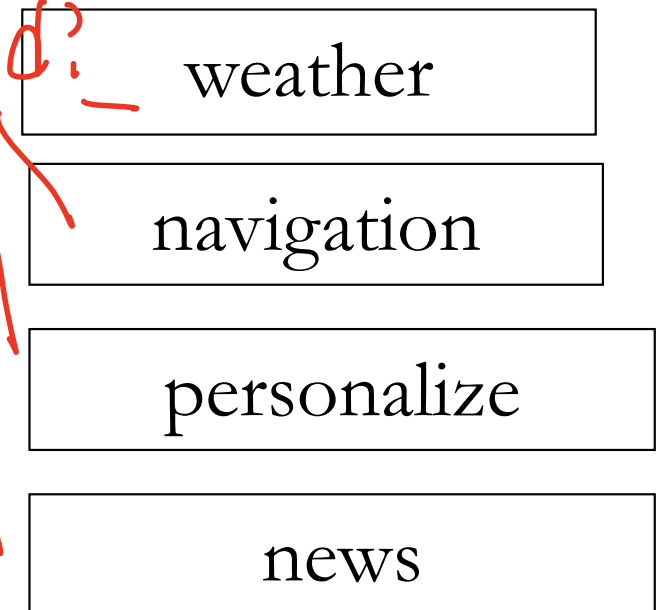
- Fragment-level caching requires establishing a template for each dynamically generated page.
 - The **template** file specifies the contents and layout of the page.
 - Essentially, each page is **factored** into a number of fragments of differing cacheability profiles and **different caching expiration time**.
 - The fragments and the templates of the pages are maintained as separate elements on the origin servers and the proxy servers.
 - A web page is **assembled at proxy caching server** when the page is requested.
 - Popularized by Akamai as part of the Edge Side Includes (ESI) initiative

template page



check valued?

tell
not
valued



Edge Side Includes (ESI)

- Edge Side Includes (ESI) is a simple markup language used to define web page fragments of differing cacheability profiles and different caching expiration time.
- ESI defines some tags that can be used in the template files.
- A template file is a normal HTML file with ESI tags.
 - Amongst the ESI tags, “*esi: include*” is used to include a fragment.

<html>

<body>

<esi:include src=http://example.com/weather.jsp />

<esi:include src=http://example.com/navigation.jsp />

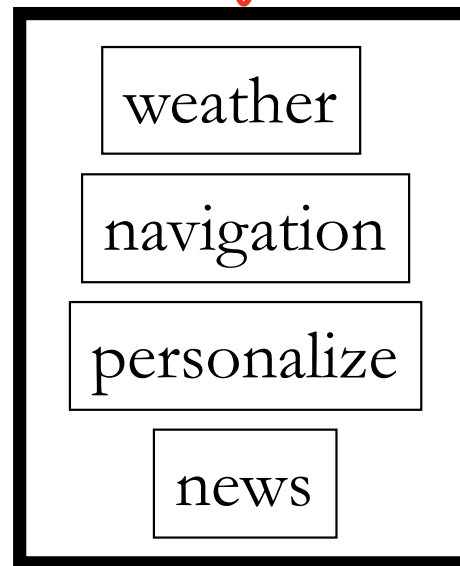
<esi:include src=http://example.com/personalize.jsp />

<esi:include src=http://example.com/news.jsp />

</body>

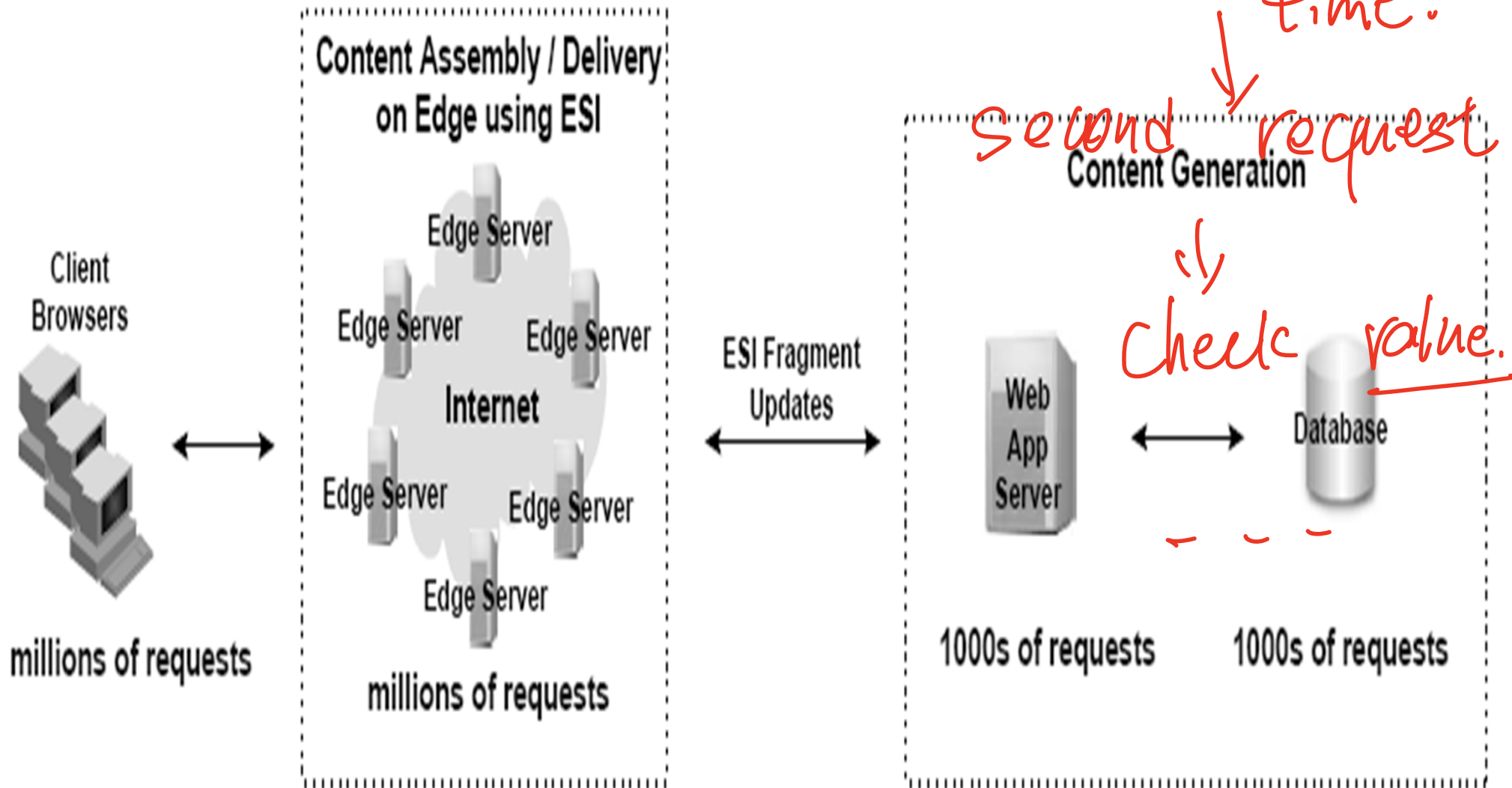
</html>

only edge server can understand



send request
to server.
return html
to client.

Save in edge
server.



- Compared with page-level caching approach, fragment-level caching approach can improve hit rates as it can specify granular caching capability. 提高了命中率

- According to the properties of different fragments, the fragment can be set with different caching expiring time and cacheability.

- problem in Fragment-level caching need the developer to partition the pages. 需分区来写

- Might require the re-writing of pages or applications

- Further reading

- https://docs.oracle.com/cd/B14099_19/caching.1012/b14046/esi.htm

Review

- What are the main steps in accessing a web applications?
- What measures can be used to reduce the response time of a web-based application? What is the techniques for implementing these measures?
- What does a web cache work? What does a web cache do to reduce the response time to web applications?
- Assume that (a) two clients access a web application through two connections with different network latencies, and, (b) the difference in network latency for the two connection is x ms. Explain why the difference in response time to the two clients might be significantly higher than x ms.

Review

- How do the forward proxy and reverse proxy work?
- Understand how web caches are placed on the Internet and how they are used.
- How does a CDN work?
- What are the problems associated with the use of web cache?
- What are dynamic pages and what are the features of the dynamic pages?
- Understand how page-level caching and fragment-level caching work and the potential problems associated with them.